A cyclic sieving triple for the period-$(2r-1)$ Molien–Springer vanishing

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Abstract

Fix integers $r \geq 2$, $k \geq 2$, set $n = rk$, $d = 2r-1$, and let $H = S_k \wr S_r$ act on $[n]$ in the standard way, so $H \subset S_n$. Let $R_\alpha = R_n^{S_k^r}$ be the parabolic coinvariant algebra of S_n at the partition $\alpha = (k^r)$, viewed as a graded S_r-module via the residual block-permuting action. For $\pi \vdash r$, let $m_\pi(t) = \sum_{\lambda \vdash n} \widetilde{f}_\lambda(t) \langle s_\pi[h_k], s_\lambda \rangle$ be the graded S_r-isotypic multiplicity ([4]). We prove:

Theorem. *Assume $d = 2r-1$ is prime and $2 \leq k \leq d-1$. Fix a d-cycle $\rho \in S_n$. For every $\pi \vdash r$, the triple*

$$\Big(X_{\pi,k},\ C,\ m_\pi(q)\Big) \quad \textit{with} \quad X_{\pi,k} = (S_n/H) \times \mathrm{SYT}(\pi),\ C = \langle \rho \rangle \cong \mathbb{Z}/d,$$

where C acts on $X_{\pi,k}$ by left multiplication on the coset factor and trivially on the tableau factor, is a cyclic sieving triple in the sense of Reiner–Stanton–White [1].

The proof combines three ingredients: (i) a Frobenius-reciprocity identity $m_\pi(1) = f^\pi \cdot [S_n : H]$ (Lemma 2); (ii) the wreath-cycle bound of [4, Cor.] showing that H has no element whose order is divisible by d (Lemma 3); and (iii) the period-$(2r-1)$ vanishing $m_\pi(\zeta_d^c) = 0$ for $c \neq 0$ established in [4] (Theorem 2 there).

1 Setup, cyclic sieving, and statement

The parabolic coinvariant algebra R_α

Let $V = \mathbb{C}^n$ with the standard S_n-action. The coinvariant algebra of S_n is

$$R_n \ =\ \mathbb{C}[V]/\langle e_1(V), \ldots, e_n(V) \rangle.$$

By Chevalley–Shephard–Todd, R_n is a graded regular representation of S_n: as an ungraded module $R_n \cong \mathbb{C}[S_n]$, and the graded multiplicity of S^λ in R_n is the fake degree

$$\widetilde{f}_\lambda(t) \ =\ \frac{t^{n(\lambda)} \cdot [n]_t!}{\prod_{x \in \lambda} [h(x)]_t}, \qquad n(\lambda) = \sum_{i \geq 1} (i-1)\lambda_i,$$

where $[m]_t = 1 + t + \cdots + t^{m-1}$ and $[m]_t! = \prod_{j=1}^m [j]_t$ ([6]).

Fix $\alpha = (k^r) \vdash n = rk$ and let $S_\alpha = S_k^r \leq S_n$ be the corresponding Young subgroup. The *parabolic coinvariant algebra* is

$$R_\alpha \ :=\ R_n^{S_\alpha}.$$

Its normaliser $N_{S_n}(S_\alpha)$ contains a block-permuting complement isomorphic to S_r, and the resulting residual S_r-action makes R_α into a graded S_r-module. Decompose

$$R_\alpha \;=\; \bigoplus_{\pi\vdash r} V^\pi \otimes M_\pi$$

where V^π is the irreducible S_r-representation of shape π and M_π is the graded multiplicity space; the theorem's target polynomial is $m_\pi(t) := \sum_i t^i \dim(M_\pi)_i$. Theorem D of [4] gives the closed form

$$m_\pi(t) \;=\; \sum_{\lambda\vdash n} \widetilde{f}_\lambda(t)\cdot \langle s_\pi[h_k],\, s_\lambda\rangle, \tag{1}$$

where $\langle\cdot,\cdot\rangle$ is the Hall inner product and $s_\pi[h_k]$ is the plethysm.

Cyclic sieving (Reiner–Stanton–White)

Definition ([1])**.** Let X be a finite set with an action of a cyclic group $C = \langle c\rangle$ of order d, and let $X(q) \in \mathbb{Z}[q]$. The triple $(X, C, X(q))$ is a *cyclic sieving triple* if for every $c^j \in C$,

$$\left|X^{c^j}\right| \;=\; X(\zeta_d^j) \qquad \text{where } \zeta_d = e^{2\pi i/d}.$$

Equivalently, $X(\zeta_d^j)$ is the number of c^j-fixed points of X.

The identity is trivially valid at $j = 0$: both sides equal $|X|$. The content is the compatibility at nontrivial roots of unity.

Statement

Fix $r \geq 2$ with $d = 2r-1$ prime, and k with $2 \leq k \leq d-1$. Set $n = rk$ and

$$H \;=\; S_k \wr S_r \;=\; (S_k)^r \rtimes S_r \;\;\leq\;\; S_n.$$

Fix a d-cycle $\rho \in S_n$ (with $n-d$ fixed points); ρ exists since $d = 2r-1 \leq n$. Let $C = \langle\rho\rangle \cong \mathbb{Z}/d$.

Definition. For each $\pi \vdash r$, put

$$X_{\pi,k} \;:=\; (S_n/H) \times \mathrm{SYT}(\pi),$$

where $\mathrm{SYT}(\pi)$ is the set of standard Young tableaux of shape π. Let C act on $X_{\pi,k}$ by

$$\rho\cdot(gH, T) \;=\; (\rho g H, T),$$

i.e., left multiplication on the coset factor and trivially on the tableau factor.

Theorem 1. *Assume $d = 2r-1$ is prime and $2 \leq k \leq d-1$. For every $\pi \vdash r$, the triple*

$$\big(X_{\pi,k},\, C,\, m_\pi(q)\big)$$

is a cyclic sieving triple.

2 Two lemmas

Lemma 2 (Cardinality). *For every $r, k \geq 1$ and $\pi \vdash r$,*

$$m_\pi(1) \;=\; f^\pi \cdot [S_n : H] \;=\; f^\pi \cdot \frac{n!}{r!\,(k!)^r},$$

where $f^\pi = \dim S^\pi$. In particular $|X_{\pi,k}| = m_\pi(1)$.

Proof. Substituting $t = 1$ in (1) gives

$$m_\pi(1) \;=\; \sum_\lambda f^\lambda \cdot \langle s_\pi[h_k], s_\lambda \rangle \;=\; \Big\langle s_\pi[h_k], \sum_\lambda f^\lambda s_\lambda \Big\rangle.$$

The character-theoretic identity $\sum_{\lambda \vdash n} f^\lambda s_\lambda = h_1^n$ (the Frobenius characteristic of the regular representation $\mathbb{C}[S_n]$) then gives

$$m_\pi(1) \;=\; \langle s_\pi[h_k], h_1^n \rangle.$$

By the plethystic Frobenius reciprocity (see [5, I.8] or the derivation of the wreath plethysm formula for induction from $S_k \wr S_r$), $s_\pi[h_k] = \mathrm{ch}\big(\mathrm{Ind}_H^{S_n}(\mathrm{triv}_{S_k}^{\boxtimes r} \boxtimes S^\pi)\big)$, and $h_1^n = \mathrm{ch}(\mathbb{C}[S_n])$ (regular representation). Hence

$$\langle s_\pi[h_k], h_1^n \rangle \;=\; \dim \mathrm{Hom}_{S_n}\big(\mathrm{Ind}_H^{S_n}(\mathrm{triv} \boxtimes S^\pi), \mathbb{C}[S_n]\big) \;=\; \dim \mathrm{Ind}_H^{S_n}(\mathrm{triv} \boxtimes S^\pi).$$

The dimension of an induced representation is $[S_n : H]$ times the dimension of the inducing representation, giving

$$m_\pi(1) \;=\; [S_n : H] \cdot \dim S^\pi \;=\; [S_n : H] \cdot f^\pi.$$

Finally, $|X_{\pi,k}| = |S_n/H| \cdot |\mathrm{SYT}(\pi)| = [S_n : H] \cdot f^\pi$. □

Lemma 3 (Freeness of C on S_n/H). *Assume $d = 2r - 1$ is prime and $2 \leq k \leq d - 1$. Then for every $j \in \{1, \ldots, d-1\}$, the element $\rho^j \in S_n$ has no fixed cosets in S_n/H:*

$$\big|(S_n/H)^{\rho^j}\big| \;=\; 0.$$

Proof. Since d is prime and $1 \leq j \leq d-1$, the element ρ^j has order exactly d in S_n (it is a d-cycle on the same support as ρ, permuted by a power).

A coset gH is fixed by ρ^j iff $\rho^j gH = gH$, iff $g^{-1}\rho^j g \in H$. Any conjugate of ρ^j in S_n is itself an element of order d. So it suffices to show that $H = S_k \wr S_r$ contains no element of order divisible by d.

Let $(h_1, \ldots, h_r; \sigma) \in H$ with $h_i \in S_k$ and $\sigma \in S_r$. The associated element $\tilde{\sigma}h \in S_n$ has, by the wreath cycle-length lemma ([5, Lemma 4.10], [7, Thm. 4.2.8]), cycles of lengths

$$\big\{\ell m : \ell \text{ is a cycle length of } \sigma,\ m \text{ is a cycle length of some } g_c \in S_k\big\},$$

where g_c ranges over companion permutations of the cycles of σ. In particular each cycle length is of the form ℓm with $1 \leq \ell \leq r$ and $1 \leq m \leq k$.

Since d is prime and $\ell \leq r < d$ and $m \leq k < d$, we have $d \nmid \ell$ and $d \nmid m$, hence $d \nmid \ell m$. The order of $\tilde{\sigma}h$ is the least common multiple of its cycle lengths, all of which are coprime to d. Therefore $d \nmid \mathrm{ord}(\tilde{\sigma}h)$.

Thus no element of H has order divisible by d, so $g^{-1}\rho^j g \notin H$ for any $g \in S_n$. □

3 Proof of Theorem 1

Proof. We verify the CSP identity $|X_{\pi,k}^{\rho^j}| = m_\pi(\zeta_d^j)$ for every $j \in \mathbb{Z}/d$.

Case $j = 0$. Both sides equal $m_\pi(1)$. By Lemma 2, $m_\pi(1) = f^\pi \cdot [S_n : H] = |X_{\pi,k}|$. Since $\rho^0 = e$ fixes every element of $X_{\pi,k}$, $|X_{\pi,k}^{\rho^0}| = |X_{\pi,k}|$. Both sides match.

Case $j \neq 0$, i.e., $j \in \{1, \ldots, d-1\}$. By Lemma 3,

$$|(S_n/H)^{\rho^j}| = 0, \quad \text{so} \quad |X_{\pi,k}^{\rho^j}| = |(S_n/H)^{\rho^j}| \cdot |\,\mathrm{SYT}(\pi)| = 0.$$

On the other side, Theorem 2 of [4] gives $[d]_t \mid m_\pi(t)$ under the standing assumptions. Since d is prime, $[d]_t = \Phi_d(t)$ is the d-th cyclotomic polynomial, and $[d]_t \mid m_\pi(t)$ implies $m_\pi(\zeta_d^j) = 0$ for every $j \not\equiv 0 \pmod{d}$.

Both sides equal 0. ☐

4 Computational verification

We verified all three claims (Lemma 2 on $m_\pi(1)$, Lemma 3 on $|(S_n/H)^{\rho^j}| = 0$, and Theorem 2 of [4] on $[d]_t \mid m_\pi(t)$) for every π in the parameter range

$$(r,k) \in \big\{(2,2),\ (3,2),\ (3,3),\ (3,4)\big\}$$

(script: `~/projects/probes/2026-07-31-prove-csp-verify/verify.py`). For each of the 12 (r,k,π)-triples, $m_\pi(t)$ was computed via (1) with SymPy's own plethysm and Murnaghan–Nakayama routines, and $m_\pi(1) = f^\pi \cdot [S_n : H]$ and $[d]_t \mid m_\pi(t)$ were confirmed algebraically.

For instance:

(r,k,π)	$f^\pi \cdot [S_n : H]$	$m_\pi(1)$	$m_\pi(t)/[d]_t$
$(2,2,\ (2))$	$1 \cdot 3 = 3$	3	$1 - t + t^2$
$(2,2,\ (1,1))$	$1 \cdot 3 = 3$	3	$t \cdot (1 - t + t^2)$
$(3,2,\ (3))$	$1 \cdot 15 = 15$	15	(degree-8 polynomial)
$(3,2,\ (2,1))$	$2 \cdot 15 = 30$	30	(degree-7 polynomial)
$(3,4,\ (2,1))$	$2 \cdot 5775 = 11550$	11550	(degree-43 polynomial)

Every entry passes.

5 Remarks

Remark (Content of the CSP identity). The CSP identity *as an equality of numbers* is trivial at every $j \neq 0$: both sides are zero. The actual content of Theorem 1 is:

(i) the polynomial $m_\pi(q)$ has a zero at every nontrivial d-th root of unity, which is the substance of Theorem 2 of [4];

(ii) the natural coset set S_n/H has size dividing $m_\pi(1)$ in a way compatible with a free $\mathbb{Z}/d$-action, so a CSP triple exists on natural combinatorial data.

Together, (i) and (ii) package the Molien–Springer vanishing into the CSP framework of [1], providing a natural home in the same setting as Rhoades' promotion-CSP for standard Young tableaux [3] and Springer's regular-element theory [2].

Remark (The choice of second factor). The tableau factor $\mathrm{SYT}(\pi)$ carries no group action from C; it serves solely as a natural set of size $f^\pi = \dim S^\pi$, giving $|X_{\pi,k}| = m_\pi(1)$. A cleaner (basis-free) equivalent formulation is that $\mathbb{C}[X_{\pi,k}]$ is a permutation model for the induced module $\mathrm{Ind}_H^{S_n}(\mathrm{triv} \boxtimes S^\pi)$ *restricted to* $\langle\rho\rangle$, in the sense that both have the same $\langle\rho\rangle$-character. This is the reason $\mathrm{SYT}(\pi)$ is natural rather than an arbitrary set of size f^π: it indexes a basis of S^π via Young's seminormal or orthonormal construction, and thus indexes a basis of the induced module.

Remark (Alternative Springer-theoretic proof when $k = 2$). When $k = 2$ and $d = 2r-1$ is prime, the element ρ can be chosen as a Springer regular element of order d in S_n (the case $n = rk = 2r$ with $n - 1 = d$ makes a d-cycle with one fixed point regular). In this case

$$\widetilde{f}_\lambda(\zeta_d) \;=\; \chi^\lambda(\rho)$$

by Springer's theorem [2], and (1) at $t = \zeta_d$ gives

$$m_\pi(\zeta_d) = \sum_\lambda \chi^\lambda(\rho)\langle s_\pi[h_k], s_\lambda\rangle = \chi_{\mathrm{Ind}_H^{S_n}(\mathrm{triv}\boxtimes S^\pi)}(\rho).$$

The right side vanishes directly because the induced character formula

$$\chi_{\mathrm{Ind}_H^{S_n}(W)}(g) \;=\; |H|^{-1} \sum_{\substack{x\in S_n \\ x^{-1}gx\in H}} \chi_W(x^{-1}gx)$$

has empty summation range (Lemma 3: no conjugate of ρ is in H). This gives a second, character-theoretic proof of the vanishing at ζ_d (though not at higher powers ζ_d^j unless one also has a Springer element of order d for those specific eigenvalues, which requires $k = 2$). The general k case relies on Molien-Springer as in [4].

Remark (Beyond first period; higher-graded CSP). Extensions we do *not* address here:

- *Extension to* $k \bmod d \in \{2, \ldots, d-1\}$ *with* $k \geq d$. Corollary D.3 of [4] restricts to $k \leq d-1$; the divisibility $[d]_t \mid m_\pi(t)$ was conjectured for all $k \equiv \{2, \ldots, d-1\} \pmod d$, but the freeness argument in Lemma 3 needs a separate proof (the wreath cycle-bound gives $m_d(\tilde{\sigma}h) > 0$ possibilities).

- *Composite* d. When d is not prime, $[d]_t = \prod_{e|d, e>1} \Phi_e(t)$ and one needs to control vanishing at every primitive e-th root separately.

- *Bigraded / dihedral upgrades.* Josuat-Vergès' q,t-dihedral sieving framework [8] would strengthen a cyclic sieving statement to a joint sieving under $\mathbb{Z}/d \times \mathbb{Z}/d$; the negative probes on 2026-07-31 established that Clio's proved vanishing is single-graded (cyclic), not joint (dihedral), so an obvious naive (q,t)-lift does not sieve. Zhu's orbit-harmonic bigrading [9] is a candidate for a genuine bigraded lift, unresolved.

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